

Why do people smoke ?

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Data Science Trainee
Springboard

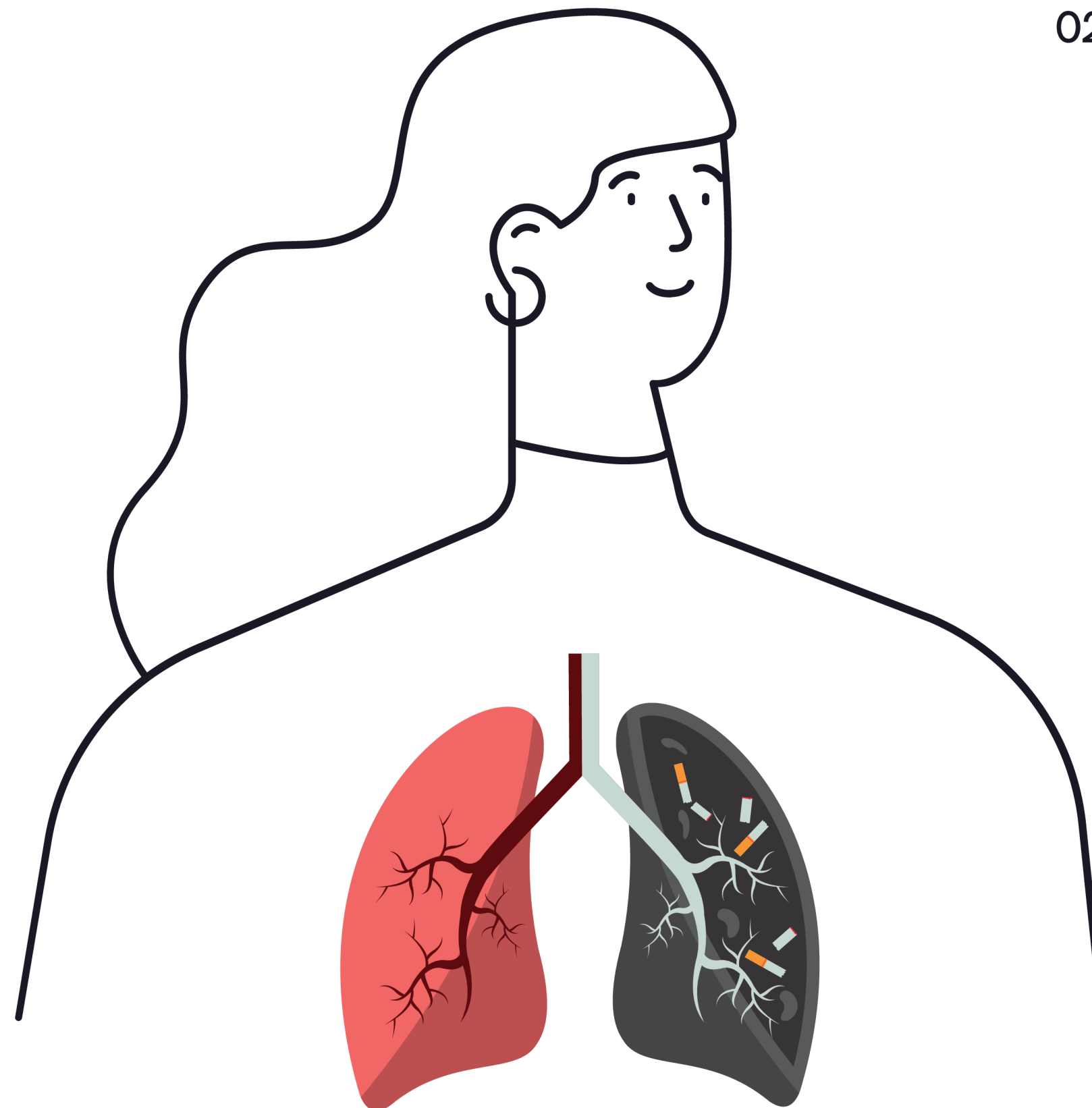


Why do we care ?

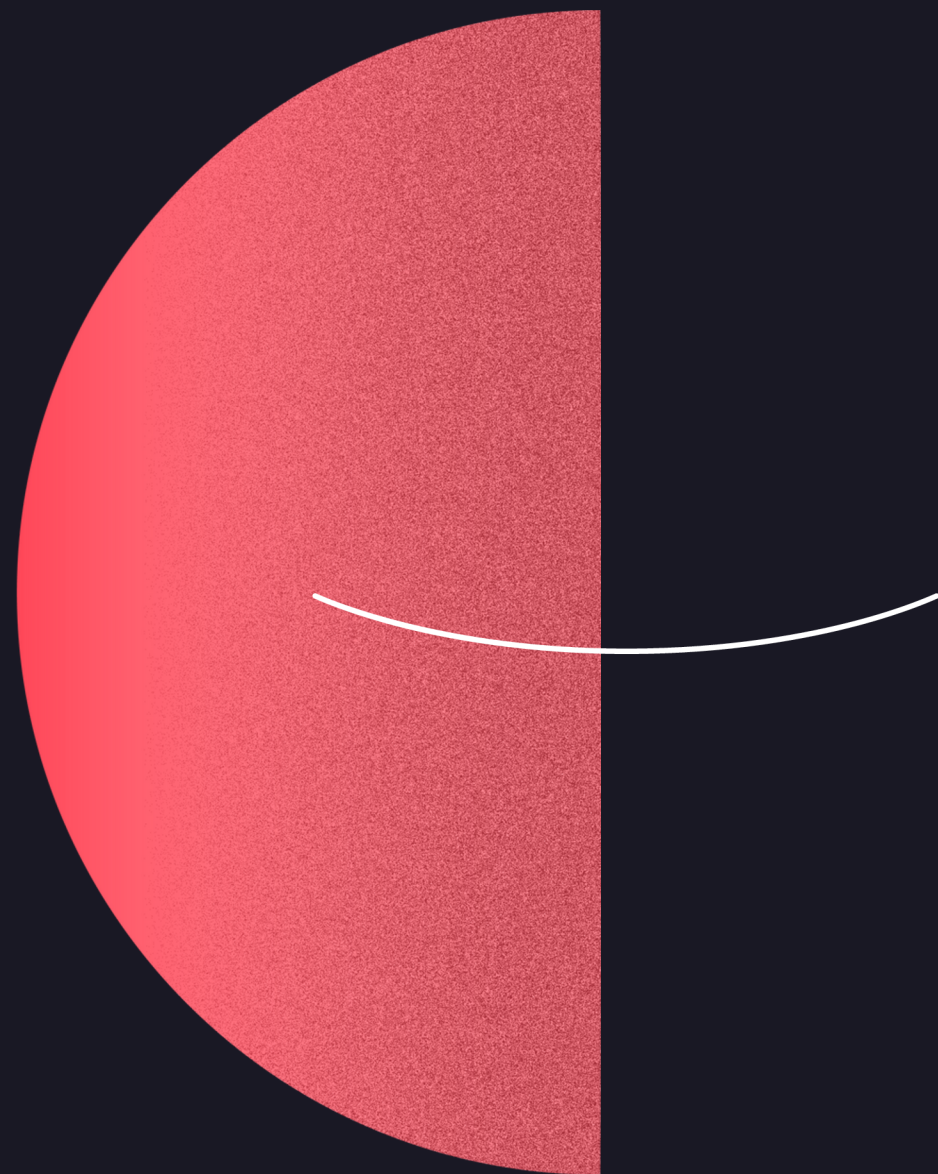
02

N 2023,
GLOBALLY **~7 MILLION** LIVES WERE LOST
USA **~500,000** LIVES

POLICY INITIATIVES ARE EVERYWHERE , BUT
ARE THEY REALLY **WORKING ??**



What we want to answer ?



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○ ○ ○ ○
○ ○ ○ ○
○ ○ ○ ○

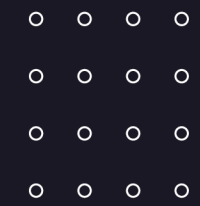
What constructs defined by Health Belief Model (HBM) contribute most towards the habit of smoking ? Are there any policy implications?

○ ○ ○ ○
○ ○ ○ ○
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○ ○ ○ ○

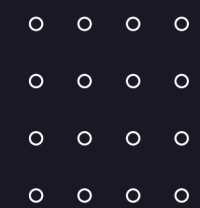
Does this trend vary by age group ?

DATA

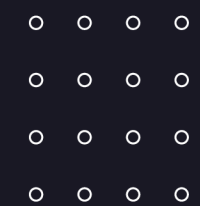
04



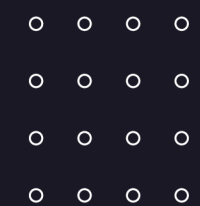
SOURCE: KAGGLE API



DATASET: BEHAVIORAL RISK
FACTOR SURVEILLANCE SYSTEM
(BRFSS)

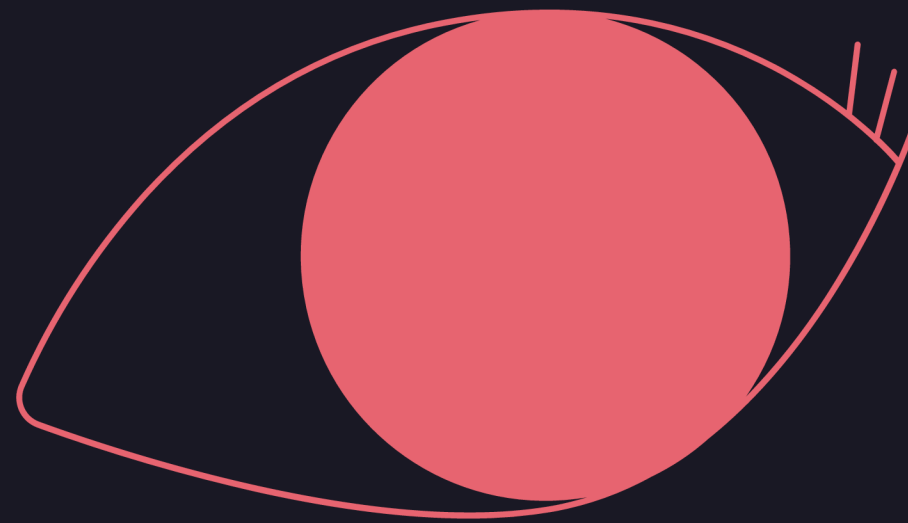


SIZE: ~ 2MILLION RECORDS AND
645 COLUMNS

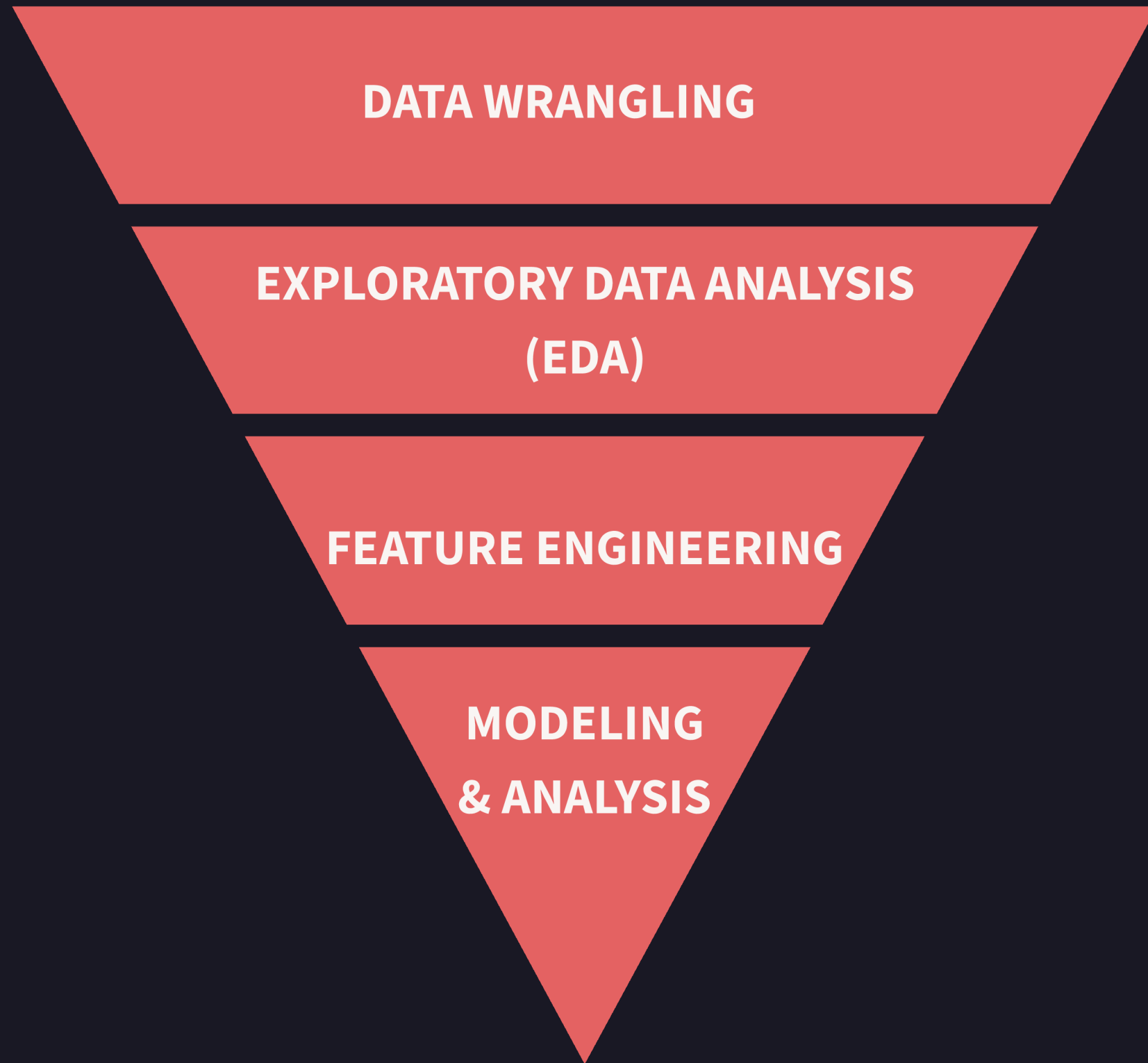


DATA TYPE: SURVEY





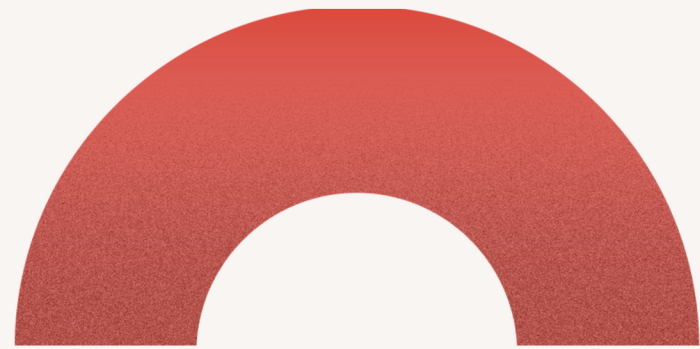
METHODOLOGY



PREDICTION MODEL

HEALTH BELIEF MODEL





DATA WRANGLING

MISSING DATA

column: Drop
0.75 threshold
row: impute

DUPLICATE FEATURES

Drop redundant
features

IMPUTE FEATURE VALUES

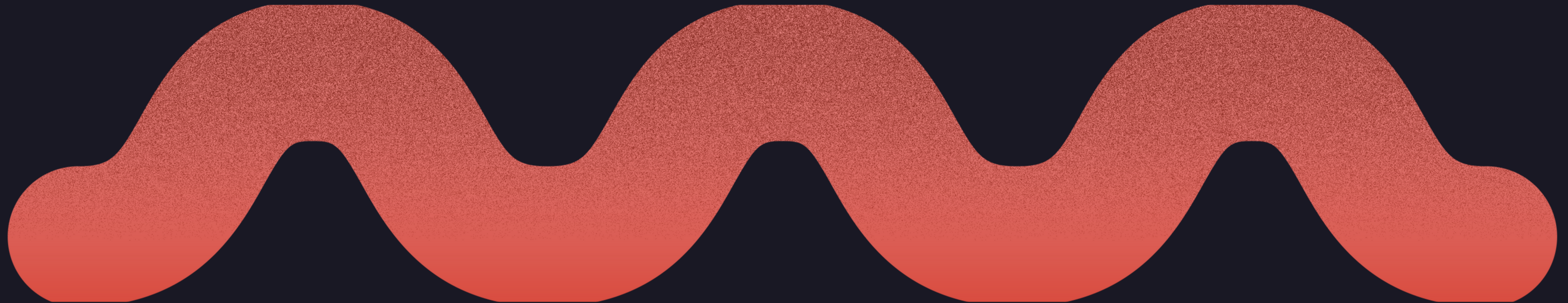
Standardize feature
values to same units

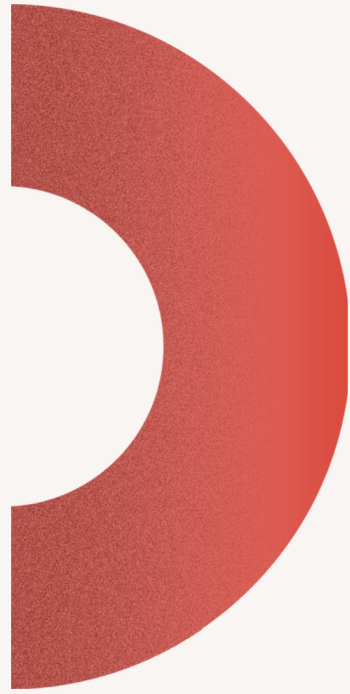
NON-INFORMATIVE FEATURES

Drop record identifiers

WRONG DATATYPES

Change datatypes
appropriately





EDA

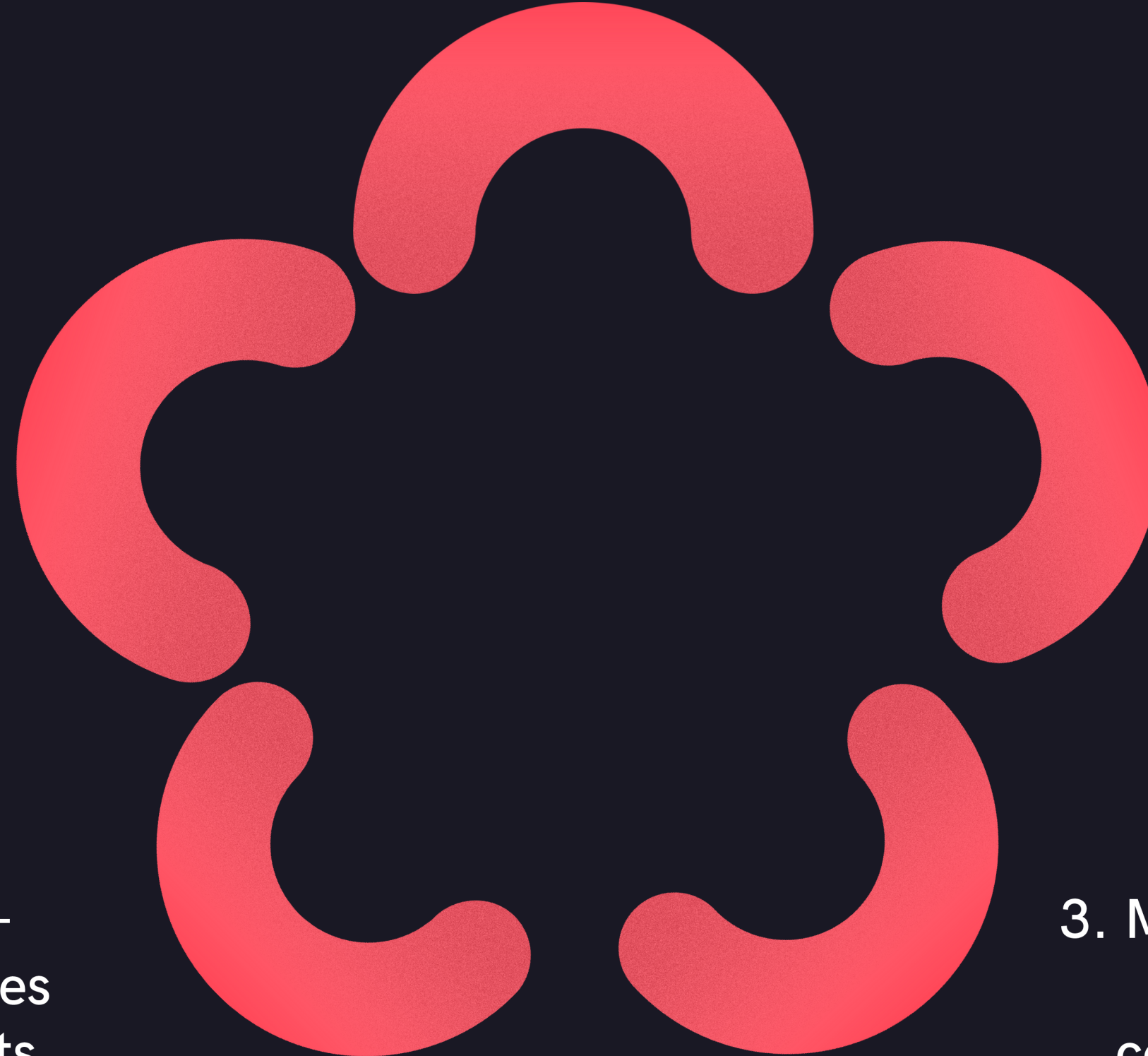
1. Selection of Target Variable

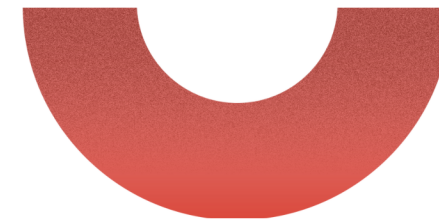
2. Data Validation

3. Map dataset
to HBM
constructs

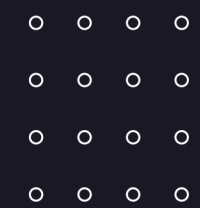
4. Binning zero-
imbalance features
into fewer buckets

5. Evaluating Data
Leakage

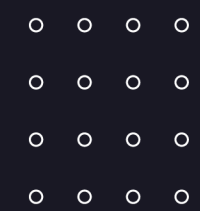




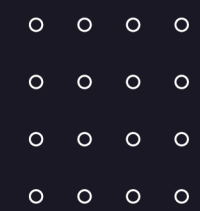
FEATURE ENGINEERING



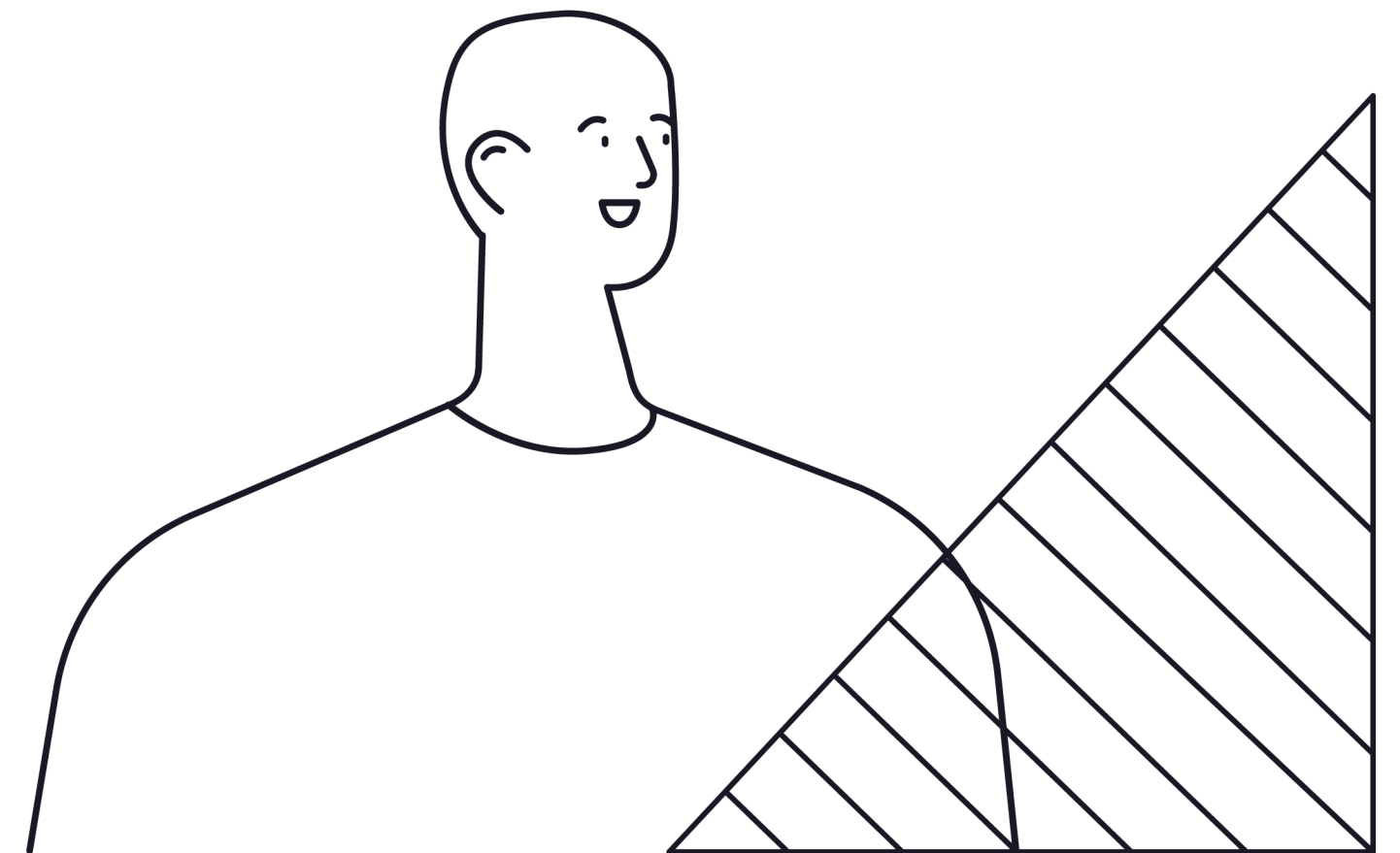
CATEGORICAL VARIABLES:
One-hot Encode

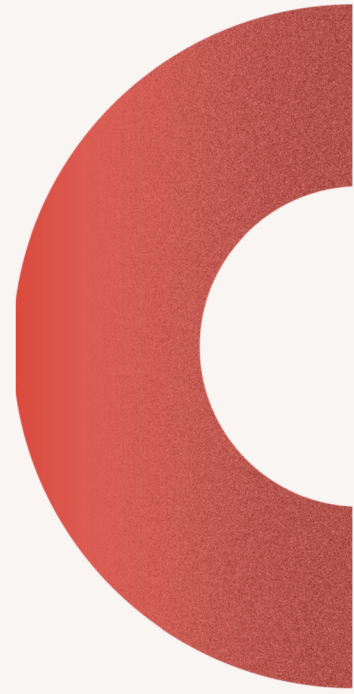


NUMERIC VARIABLES:
Behavioral Risk Factor
Surveillance System (BRFSS)



GENERATE TRAIN/TEST SPLIT:
~1.9 million records for Train set
~500,000 records for Test set





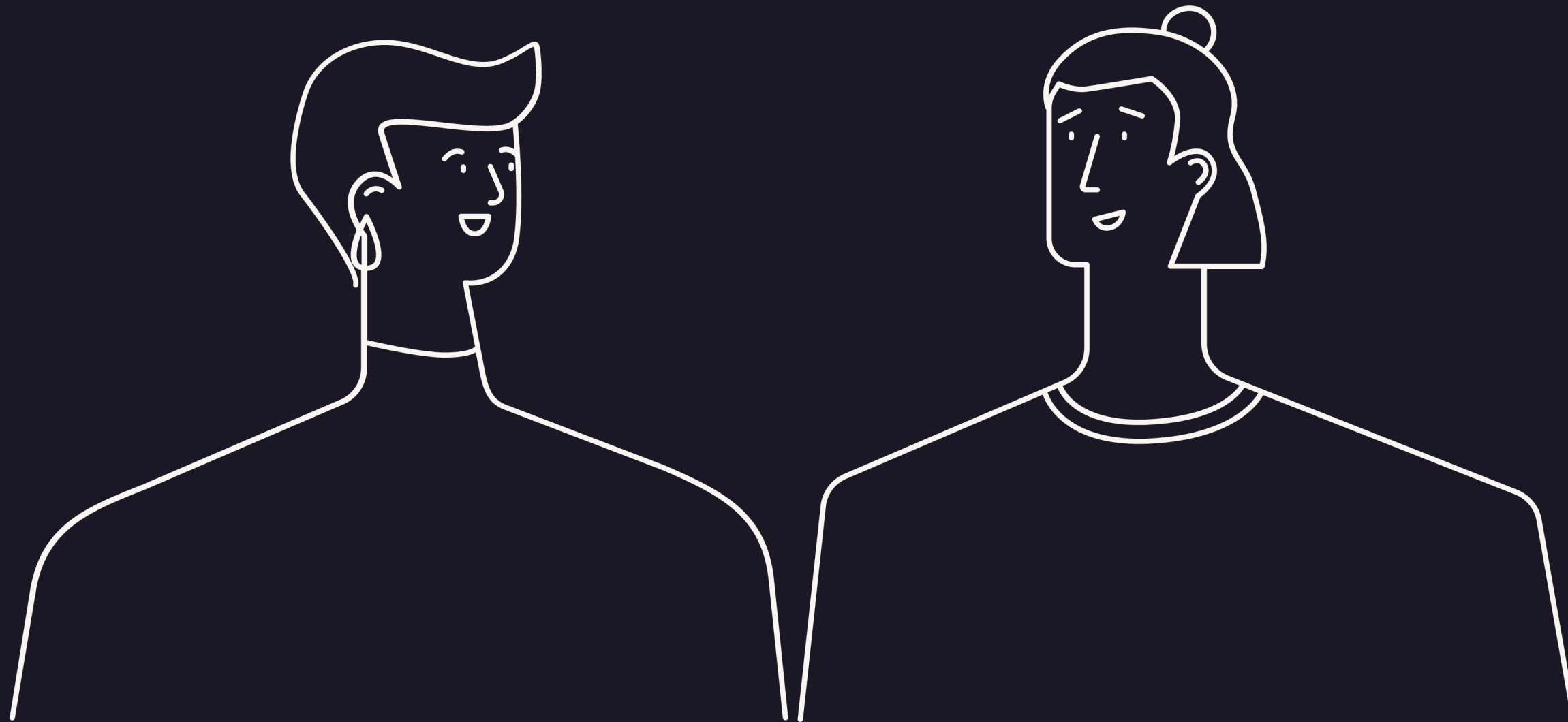
MODELING

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC	Training Time
CatBoost	0.729721	0.333598	0.713685	0.454670	0.796675	0.458032	1144.751788
CatBoost Tuned	0.730858	0.334517	0.712342	0.455249	0.796907	0.458349	1831.023640
LightGBM	0.728925	0.332754	0.713301	0.453807	0.795863	0.456280	16.839819
LightGBM Tuned	0.730335	0.333869	0.711520	0.454481	0.796365	0.457688	3298.030455

ANALYSIS

What constructs defined by Health Belief Model (HBM) contribute most towards the habit of smoking ? What are some policy implications ?

Does this trend vary by age group ?



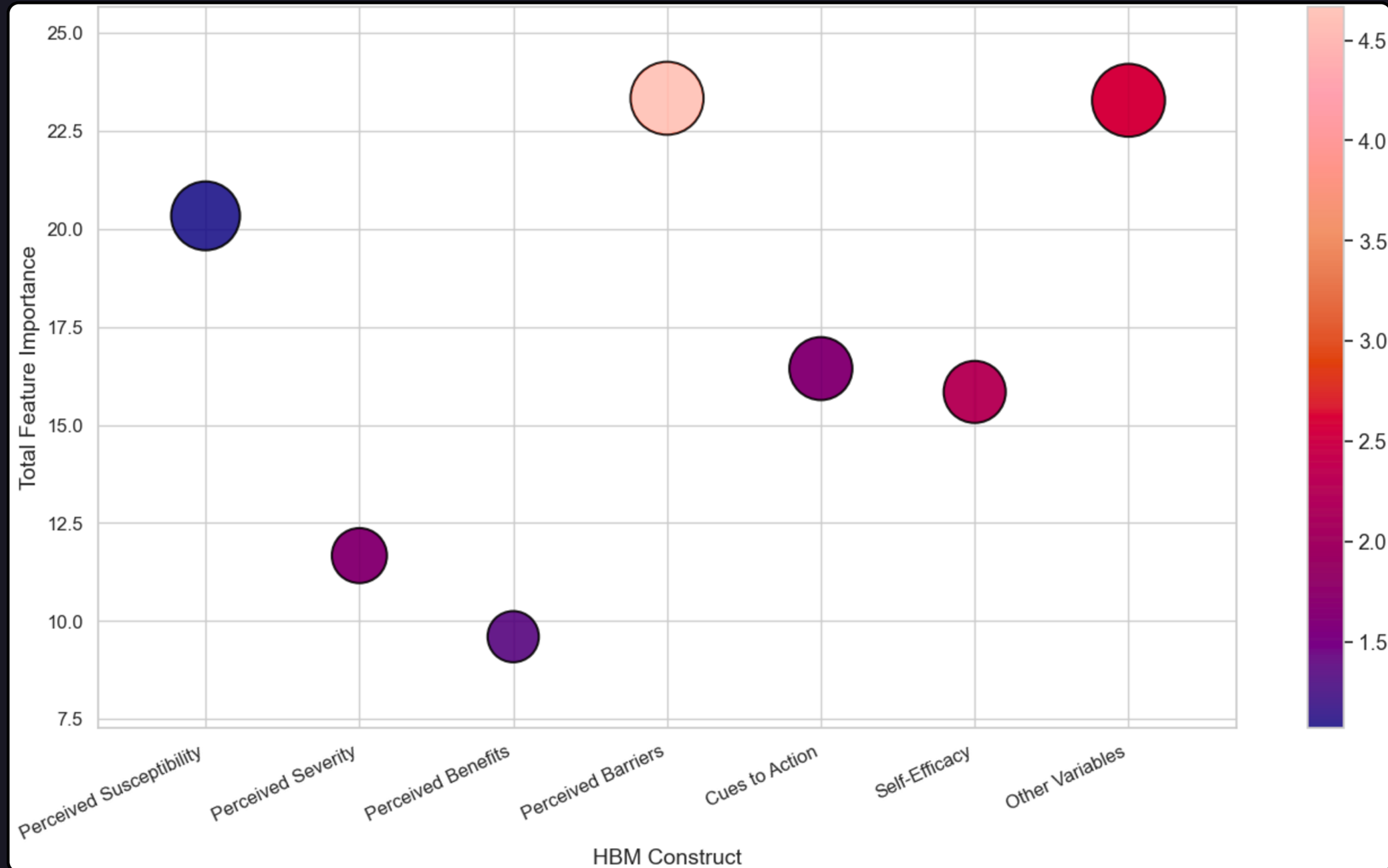
KEY INSIGHTS

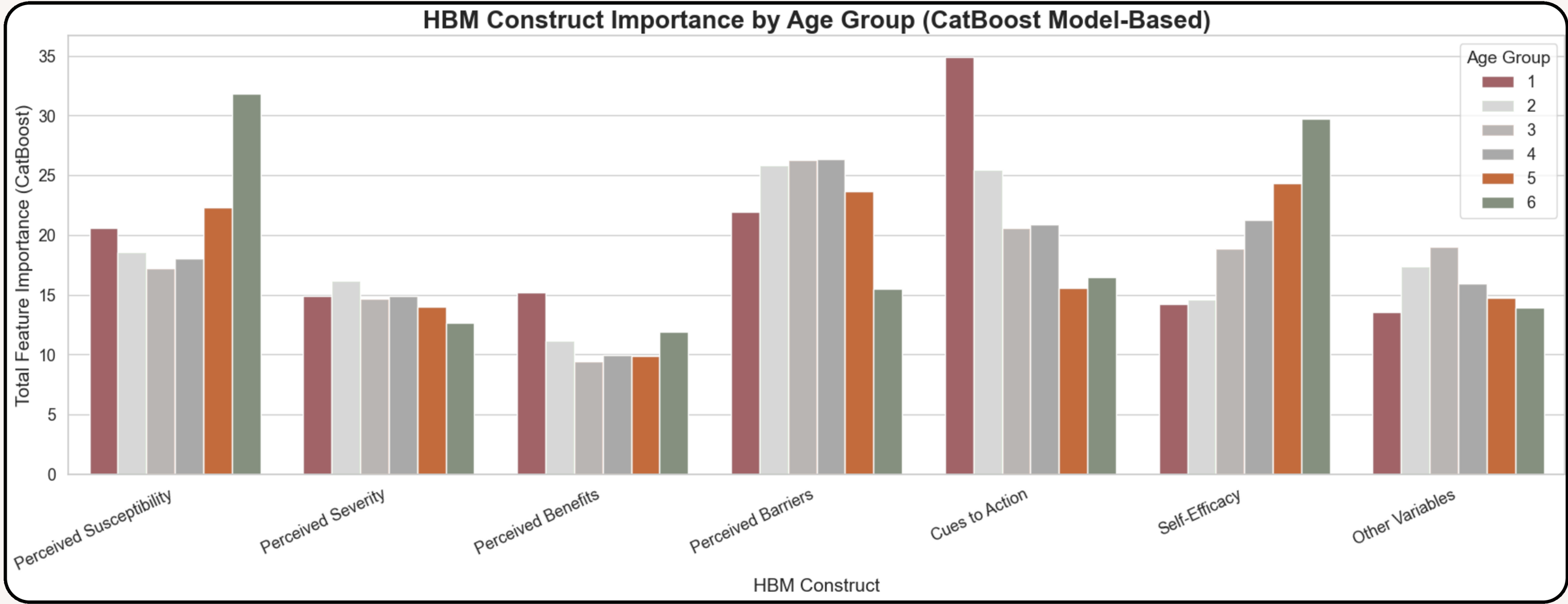
Perceived Barriers (**Education, Income**) are the most actionable levers for intervention

Demographics (**age, marital status**) remain foundational

Cues to Action like (**binge drinking, routine checkups**) signal opportune moments for targeted messaging.

Self-efficacy indicators (**weight, activity**) matter more than perceived severity.





Code	Age Range	Description
1	18–24	Young Adults
2	25–34	Early Career Adults
3	35–44	Mid-Career Adults
4	45–54	Mature Adults
5	55–64	Pre-Retirees
6	65+	Seniors

18-24

Most influenced by **Cues to Action** and Perceived Benefits

Leverage peer influence, social media, benefits messaging.

25-54

Dominated by **Barriers**.

Address barriers: stress, time, cost, support access.

55-64

Influenced most by **Susceptibility**.

Emphasize risk and susceptibility via health condition links.

65+

High on **Self-Efficacy**, **Susceptibility**, and **Benefits**.

Empower with tools and reinforce confidence to quit.



Future work

I would also like to add more recent data to my project (~2021-2024) to evaluate differences in smoking habits and differences in constructs over a decade.

